

PageRank Algorithm

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1 Theory behind the method

The purpose of the PageRank method is to generate a ranking of the pages in a web, in order to show them sorted by importance in response to a web search.

To order these pages, the main idea is to consider the links connecting one page to the other. Normally a page that is linked from most of all the other pages should be ranked higher than a page that is not linked by others. To be more consistent, it is relevant to weight the contributions of each page consider the total number of outgoing links.

Suppose that the web is composed of n pages. Denote by x_j for $j \in \{1, \dots, n\}$ the score of the page j , n_j the number of links going out from page j and L_k the set of indexes of the pages that links to page j , for a given page k we can compute its score by using the following formula:

$$x_k = \sum_{j \in L_k} \frac{x_j}{n_j} \quad (1)$$

The 1 equation, considered for all the pages, generates a linear system that can be rewritten as

$$A\mathbf{x} = \mathbf{x}, \quad (2)$$

where $A \in \mathbb{R}^{n \times n}$ is the coefficient matrix and $\mathbf{x} \in \mathbb{R}^n$ the vector containing all the pages.

Definition 1.1. Let $M \in \mathbb{C}^{n \times n}$ be a matrix. $\lambda \in \mathbb{C}$ is called an *eigenvalue* of M , if exists a vector $\mathbf{v} \in \mathbb{C}^n$, $\mathbf{v} \neq 0$ such that $M\mathbf{v} = \lambda\mathbf{v}$. The pair $(\lambda, \mathbf{v})$ is called an *eigenpair* of M .

Definition 1.2. Let $M \in \mathbb{R}^{n \times n}$ be a matrix with nonnegative entries. M is *column-substochastic* if the sum of all its entries in a columns is less than or equal to 1. When the sum of all its entries in each column sum up to 1, M is called *column-stochastic*.

Proposition 1.1. Every column-stochastic matrix has 1 as an eigenvalue

Proof. Let $A \in \mathbb{R}^{n \times n}$ be a column-stochastic matrix, $\mathbf{e}$ a column vector with all entries equal to 1 and $A^\top$ the transpose matrix of A . $A^\top \mathbf{e} = \mathbf{e}$ because the row-column yields a vector of one. Since a matrix and its transpose have the same eigenvalues, then 1 is an eigenvalue of A . $\square$

From proposition 1.1, $(1, \mathbf{x})$ is an eigenpair of matrix A (definition 1.1) if A is column-stochastic (definition 1.2).

Thus, the ranking problem can be reduced to the search of the eigenvector of A that correspond to the eigenvalue $\lambda = 1$.

The condition on the column-stochasticity of A is not already given: in order to compute the formula in equation 1 and be aware that A is column-stochastic, n_j should be different than 0. When $n_j = 0$, the page j is called a dangling node and the all the entries in column j are zeros. In this case A is column-substochastic.

In the case of A column-substochastic the spectral radius $\rho(A)$ is less than or equal to one (as shown in proposition 1.2), thus it is not guaranteed that 1 is an eigenvalue. So the problem becomes more difficult to handle, but there are several solutions that can be applied in order to make A be column-stochastic.

Proposition 1.2. Let $A \in \mathbb{R}^{n \times n}$ be a column-substochastic matrix, then $\rho(A) \leq 1$ where $\rho(A)$ is the spectral radius of A .

Proof. Since A is column-substochastic, we have $\sum_i A_{ij} \leq 1 \forall j$. Thus $A^\top$ is row-substochastic and

$$\|A^\top\|_\infty = \max_i \sum_j (A^\top)_{ij} \leq 1.$$

Since $\|A^\top\|_\infty = \|A\|_1$ and $\rho(A) \leq \|A\|_1$, it follows that $\rho(A) \leq 1$. $\square$

Another important observation is that, for a column-stochastic matrix A , the eigenspace associated with the eigenvalue 1, denoted by $V_1(A)$, has to be of dimension 1 in order to have a unique ranking. In this case, there exists a unique eigenvector $\mathbf{x}$ with $\sum_i x_i = 1$ associated with eigenvalue 1, that can be used as the ranking vector.

The condition $\dim(V_1(A)) = 1$, holds if and only if the directed graph associated with A is strongly connected (A positive and column-stochastic). Otherwise, it is possible to represent A by a block diagonal structure

$$A = \begin{bmatrix} A_1 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & A_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & A_r \end{bmatrix}$$

where A_i for $i \in \{1, \dots, r\}$ is the sub-matrix associated with the i component over the r components of the matrix A . Each matrix A_i is column-stochastic and admits an eigenvector $\mathbf{v}_i$ with eigenvalue 1. By constructing the vectors $\mathbf{w}_i$ in the following way:

$$\mathbf{w}_1 = \begin{bmatrix} \mathbf{v}_1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{w}_2 = \begin{bmatrix} 0 \\ \mathbf{v}_2 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots$$

where the zeros are for all the elements corresponding to other blocks than block i . The set of $\mathbf{w}_i$ for $i \in \{1, \dots, r\}$ are linearly independent eigenvectors for A with eigenvalue 1. Thus $V_1(A)$ has dimension at least r .

In order to generate unambiguous score vector, therefore having $\dim(V_1(A)) = 1$, it is possible to modify the matrix A in the following way: suppose $A \in \mathbb{R}^{n \times n}$ to be a column-stochastic matrix (hence there are no dangling nodes), $S \in \mathbb{R}^{n \times n}$ be column-stochastic matrix with all entries $1/n$ and $m \in [0, 1]$, then

$$M = (1 - m)A + mS \quad (3)$$

is a matrix of a weighted average of A and S . The number $1 - m$ is called damping factor and normally its value is 0.85, so $m = 0.15$.

Before presenting a crucial theorem for the PageRank algorithm, there are a useful definition and theorem.

Definition 1.3. Let A be a matrix. A is called *reducible* if there exists a permutation matrix P such that $P^\top A P$ is block upper triangular, i.e.

$$P^\top A P = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix},$$

where A_{11} and A_{22} are square matrices. If no such permutation matrix exists, A is called *irreducible*.

Theorem 1.1. A matrix A is irreducible if and only if its adjacency graph is strongly connected.

Theorem 1.2 (Perron-Frobenius Theorem). Let $A \in \mathbb{R}^{n \times n}$ be a matrix with non-negative entries. If A is irreducible, then each of the following is true.

- $r = \rho(A) \in \sigma(A)$ and $r > 0$.
- The algebraic multiplicity of r is 1.
- There exists an eigenvector $\mathbf{x} > \mathbf{0}$ such that $A\mathbf{x} = r\mathbf{x}$.
- The unique vector defined by $A\mathbf{p} = r\mathbf{p}$, $\mathbf{p} > \mathbf{0}$ and $\|\mathbf{p}\|_1 = 1$, is called the *Perron vector*. There are no nonnegative eigenvectors for A except for positive multiples of $\mathbf{p}$, regardless of the eigenvalue.

Since the entries of A are non-negative and the entries of S are strictly positive, the matrix $M = (1 - m)A + mS$ is a positive matrix. Therefore, the adjacency graph associated with M is strongly connected and, by Theorem 1.1, M is irreducible. Moreover, M is column-stochastic, hence 1 is an eigenvalue of M and $\rho(M) = 1$.

By the Perron-Frobenius Theorem 1.2, the eigenvalue $\lambda = 1$ is simple and admits a unique eigenvector $\mathbf{x} > \mathbf{0}$ (up to scaling). Furthermore, all other eigenvalues λ_i of M satisfy $|\lambda_i| < 1$.

Consequently, $\dim(V_1(M)) = 1$ and the ranking vector is uniquely defined by the normalization condition $\|\mathbf{x}\|_1 = 1$.

To compute the ranking eigenvector $\mathbf{x}$ of the M , it is possible to use the Power Method since M satisfies the condition of having a dominant eigenvalue λ_1 , i.e. $|\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_n|$.

At the end, we have a unique ranking given by the values of the eigenvector $\mathbf{x}$, with $\sum_i x_i = 1$.

2 PageRank algorithm

Theoretical Foundations

In the previous section, the PageRank matrix and its main properties were introduced. It is now shown why the Power Method is a suitable and effective choice for computing the ranking vector $\mathbf{x}$.

To apply the Power Method, the matrix M must have a dominant eigenvalue, which in this case is $\lambda = 1$. The Power Method is an iterative procedure that converges to the eigenvector associated with the dominant eigenvalue, up to a scaling factor. This property is essential for computing the ranking vector of the web.

The convergence of the method is guaranteed by the following theorem.

Theorem 2.1. Let $A \in \mathbb{R}^{n \times n}$ have eigenvalues $\lambda_1, \dots, \lambda_n$ with $|\lambda_1| > |\lambda_i|$ for all $i \geq 2$, and let $\mathbf{x}_1$ be an eigenvector associated with λ_1 . Consider the sequence

$$\begin{aligned} \mathbf{v}^{(0)} &= \mathbf{v} = \alpha_1 \mathbf{x}_1 + \sum_{i=2}^n \alpha_i \mathbf{x}_i, \quad \alpha_1 \neq 0 \\ \mathbf{v}^{(m+1)} &= A \mathbf{v}^{(m)}. \end{aligned}$$

Then

$$\begin{aligned} \lim_{m \rightarrow \infty} \frac{\mathbf{v}^{(m)}}{\|\mathbf{v}^{(m)}\|_2} &= (-1)^s \mathbf{x}_1, \quad s = sg(\lambda_1)sg(\alpha_1) \\ \lim_{m \rightarrow \infty} r_A(\mathbf{v}^{(m)}) &= \frac{\mathbf{v}^{(m)\top} A \mathbf{v}^{(m)}}{\|\mathbf{v}^{(m)}\|_2^2} = \lambda_1, \end{aligned}$$

where $r_A(\mathbf{v}^{(m)})$, is the Rayleigh quotient of $\mathbf{v}^{(m)}$ w.r.t A .

In the case of the PageRank using matrix M , the Power Method converges to the unique positive eigenvector $\mathbf{x}$ corresponding to the dominant eigenvalue $\lambda = 1$, in accordance with the Perron–Frobenius theorem.

Graph Representation and Preprocessing

To represent the web graph efficiently, the link structure is stored using adjacency lists that track both outgoing links and incoming links (backlinks) for each page i .

During graph preprocessing, dangling nodes (pages) with no outgoing links are identified and stored in a binary indicator vector $\mathbf{e}_{\text{dangling}}$. Additionally, the out-degree n_j of each node j is computed to properly weight the link contributions during the PageRank computation.

Sparse Matrix Representation (CSR)

To achieve computational efficiency and minimize memory usage, the link matrix A is represented using the Compressed Sparse Row (CSR) format. The non-zero entries of A , defined by the normalized link weights $1/n_k$, are stored in array $\mathbf{AA}$, their corresponding column indices in $\mathbf{JA}$, and row pointers in $\mathbf{IA}$.

This format enables a fast matrix-vector multiplication optimized for sparse structures, implemented as follows:

```
def CSR_matvec(AA, JA, IA, v):
    n = len(IA) - 1
    w = np.zeros(n)
    for i in range(n):
        for k in range(IA[i], IA[i+1]):
            w[i] += AA[k] * v[JA[k]]
    return w
```

Power Method Implementation

The ranking vector $\mathbf{v}$ corresponding to the dominant eigenvector is computed using the Power Method. To maintain $O(|E|)$ efficiency, the transition matrix $M = (1 - m)A + mS$ is never explicitly constructed. Instead, its matrix-vector product is evaluated by summing three separate components: the sparse matrix product via `CSR_matvec`, the dangling node uniform redistribution, and the teleportation factor.

The iterative procedure is implemented as follows:

```
def PowerMethod(AA, JA, IA, m, maxIter, relTol):
    inv_n = 1.0/nr_pages
    teleport = m*inv_n
    v0 = np.full(nr_pages, inv_n)

    for k in range(maxIter):
        v = (1-m)*CSR_matvec(AA, JA, IA, v0)
        v += (1-m)*inv_n*(np.dot(e_dangling_nodes, v0))
        v += teleport

        v = v/v.sum()

        if np.linalg.norm(v-v0,1)<relTol:
            return v, k
        else:
            v0 = v

    return v, k
```

Implementation Notes

Several key aspects of the implementation require further examination.

First, although the matrix $M = (1 - m)A + mS$ is theoretical foundation for guaranteeing the existence and uniqueness of the ranking vector, it is never explicitly constructed in memory. Instead, the contributions of A and S are kept separate to exploit the sparsity of A . By storing A in CSR format, its matrix-vector product is computed with high efficiency. To ensure that A is column-stochastic, dangling nodes are handled by assigning them a uniform probability. The contribution of the matrix S is then added by broadcasting its effect directly to the vector. This approach is convenient because the update of $\mathbf{v}$ consists of three additive terms, but only one of them requires an explicit matrix-vector product.

Second, since the dominant eigenvalue $\lambda_1 = 1$ is known *a priori*, no explicit estimation of λ is required (e.g., via Rayleigh quotient), avoiding unnecessary vector operations and reducing the computational cost per iteration.

Finally, the stopping criterion is based on the residual between two successive iterates rather than on the convergence of the dominant eigenvalue. In particular, the iteration terminates when the L^1 -norm of the difference $\|\mathbf{v}^{(k)} - \mathbf{v}^{(k-1)}\|_1$ falls below a predefined tolerance `relTol`. Since the power method generates a sequence converging to the desired eigenvector, a small residual implies that further iterations would only lead to minor corrections. Moreover, the use of the L^1 -norm is particularly appropriate in this context, since the PageRank vector is non-negative and normalized, and the residual can be interpreted as the total change in the PageRank values across the pages. Monitoring eigenvalue convergence alone would not guarantee that the associated eigenvector has reached a satisfactory level of accuracy.

Experimental Parameters

The algorithm relies on the following default hyperparameter values:

- Damping factor parameter: $m = 0.15$ (corresponding to a standard damping factor $1 - m = 0.85$).
- Maximum iterations: `maxIter` = 100.
- Stopping tolerance: `relTol` = 10^{-4} .

Experimental Results

The algorithm was tested on the `Hollins.dat` dataset, representing a web graph of university pages. The top 10 pages ranked by their computed PageRank score and their corresponding URLs are reported below:

Top 10 PageRank scores:

```
[0.01988523 0.00929212 0.00861454 0.00806908 0.00803046 0.00716806  
0.00658446 0.00599235 0.00557425 0.00444939]
```

Corresponding page links:

```
['http://www.hollins.edu/'  
'http://www.hollins.edu/admissions/visit/visit.htm'  
'http://www.hollins.edu/about/about_tour.htm'  
'http://www.hollins.edu/htdig/index.html'  
'http://www.hollins.edu/admissions/info-request/info-request.cfm'  
'http://www.hollins.edu/admissions/apply/apply.htm'  
'http://www.hollins.edu/academics/library/resources/web_linx.htm'  
'http://www.hollins.edu/admissions/admissions.htm'  
'http://www.hollins.edu/academics/academics.htm'  
'http://www1.hollins.edu/faculty/saloveyca/clas%20395/Sculptu
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